

Jacob Edwards, PhD

Data Scientist

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portfolio: <https://amnion.github.io/>

I have 13 years' experience using data, code and models to solve problems. I have a broad skillset with a specialty in time-series signals and other unstructured data. I know and love the challenges brought on by messy, real-world data. I am meticulous and action-oriented, and I approach data projects with the rigor and imagination that come from disciplined training in science.

SKILLS

Programming & Tools

Python, SQL, R, MATLAB, MS Office

Data Analysis

Statistical modeling, regression and classification, time series analysis, natural language processing, machine learning, feature engineering, dimensionality reduction, model evaluation and prediction, data cleaning, wrangling, and visualization

Experimentation

Hypothesis testing, experimental design, sampling methods, effect size estimation, bootstrapping, simulation

Communication

Data storytelling, technical writing, oral presentation, cross-functional collaboration, teaching and mentoring

RELEVANT EXPERIENCE

Research Scientist

COLUMBIA UNIVERSITY

New York, NY

2019 - 2025

Led end-to-end execution of research projects from hypothesis formulation and experimental design to data collection, analysis and modeling, and delivery to stakeholders as technical reports for publication or presentations at Columbia and national conferences.

Time-Series Modeling of Brain Data

- Generated and analyzed 3.2 TB of time-series neural data. Built scalable workflows for signal cleaning, feature extraction, aggregation and visualization.
- Implemented numerous analysis techniques to understand stimulus-response relationships, including kernel smoothing, adaptive thresholding for event detection, auto- and cross-correlation functions, and Fourier transforms.
- Replaced a manual data quality screening workflow with an automated, supervised ML-based [anomaly detection classifier](#), reducing manual review time by ~20% and improving consistency across projects.
- Built [interactive dashboards](#) in Python (Dash) and MATLAB to support exploratory analysis and enable non-technical collaborators to interpret multi-dimensional results.

Sequence Modeling of Vocal Acoustic Data

- Generated and analyzed large-scale acoustic datasets comprising more than 24,000 vocal events across 138 subjects. Optimized event segmentation pipeline to derive structured tabular representations from raw waveform signals.
- Engineered features from audio recordings using domain metrics and decomposition techniques (PCA, wavelets). Applied dimensionality reduction and clustering techniques to identify latent structure and feature correlations.
- Created a [graphical user interface](#) to unify database management and data entry pipelines. The GUI trivialized manual annotation (reduced work hours per project ~80%) and enabled interns to contribute to project progress.
- Developed and evaluated NLP model for sequence prediction using non-lexical signal features. Designed a custom cost metric based on sequence distance algorithms. Created custom effect size metrics and bootstrapped confidence intervals.

Leadership and Cross-Functional Impact

- Secured \$140,256 in competitive [federal research funding](#) by translating technical proposals into two articulated project aims and measurable outcomes.
- Developed and taught an original, [applied data science curriculum](#) (20 hours of lecture, 7 homework assignments) covering statistical modeling, data wrangling, visualization and machine learning. Emphasized translating real-world data analysis into actionable policy recommendations for legislators and decision-makers.
- Delivered presentations to technical and non-technical audiences via posters and talks within the department, at national conferences (*Society for Neuroscience*), and community events at *Taste of Science NYC* and *BioBus Harlem*.
- Authored 6 peer-reviewed papers in high-impact journals such as *Nature Scientific Reports* and *Journal of Neuroscience*.

SELECTED PUBLICATIONS

2026	Syntax models of species' rules accurately predict vocal sequences	[in press]
2025	Temporal organization of behavior is driven by species identity	paper
2023	Machine learning analysis of song links vocal acoustics to ancestry	paper

EDUCATION

PhD	2025	Columbia University
MSc	2018	Colorado State University
BSc	2013	Uni. of Tennessee, Knoxville